

SHARIF
UNIVERSITY OF TECHNOLOGY



Computer Structure

Pre-Lab Report 1

Aria Fateh Kerdari
403110382

May 17, 2026

1 Questions

1. An ALU (Arithmetic Logic Unit) is a core component of the CPU that performs arithmetic and logical operations on binary data. Arithmetic operations such as, addition, subtraction and increment/decrement. Logical operations such as, AND, OR, NOT, XOR and bit shifting.
2. **S0-S3** and **M** is for the function select. 16 different operation is available. Due to the "AC characteristic" table in the data sheet, the longest delay is for $\overline{A \oplus B}$.
3. Inputs are given to ALU through **A** and **B** input pins. With the function select pins we chose XNOR ($S_0 = S_3 = M = H$ $S_1 = S_2 = L$).
4. Due to the AT28C16 data sheet, it has 16k memory. Maximum address to output delay is 150 ns.
5. By connecting the synchronous counter to addresses(first 4) we can have first 16 memory locations.

2 Simulation

2.1 XOR From NAND

The designed circuit is shown in figure (1). Truth table associated with this circuit is shown below.

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

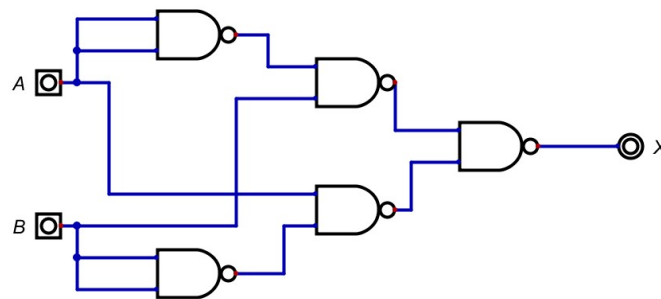


Figure 1: XOR gate is implemented with NAND .

2.2 Counting Sheep

The circuit implementation is shown in figure (2).

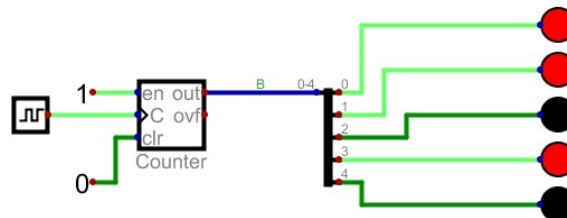


Figure 2: A Counter which counts form 0 to 31 and the number is shown with LEDs.